Mostafa Moawad Ibrahim

Mechanical Power Engineering
Kafr El-Sheikh, Egypt +20 1029362097 mostafamoawad2004@gmail.com
linkedin.com/in/mostafamoawad

Summary

Mechanical Engineer with a strong foundation in mechanical design and engineering analysis, with hands-on experience using SolidWorks, AutoCAD, and ANSYS. Skilled in 3D modeling, mechanical component design, and engineering analysis, with a strong interest in HVAC design and systems. Holds professional certifications in SolidWorks and HVAC, with strong problem-solving abilities and a passion for applying engineering knowledge to real-world projects.

Education

Faculty of Engineering , Kafr El-Sheikh University **Oct. 2022 – Jun. 2027**
B.Sc. in Mechanical Engineering — GPA: 2.5 / 4.0 *Kafr El-Sheikh, Egypt*

Technical Skills

CAD & 3D Modeling: SolidWorks (CSWA Certified), 3D CAD Modeling, xDesign, xShape, Mold Design, 2D Drafting, AutoCAD
Simulation & Analysis: ANSYS Fluent, CFD Simulation, Finite Element Analysis (FEA), Boundary Layer Meshing, Thermal-Hydraulic Performance
Manufacturing & Specializations: Additive Manufacturing (3D Printing), Sustainable Design, Electrical Design, HVAC System Design
Thermal & Fluid Engineering: Fluid Mechanics, Heat Transfer Analysis, Exergy Analysis, Thermodynamics, Internal Combustion Engines
Data Analysis & Software: Advanced Excel, Data Processing, Mathematical Modeling, Experimental Validation
Languages & Digital Media: Arabic (Native), English (Professional Working), Video Editing, Audio Editing, Technical Documentation

Experience

General Petroleum Company (GPC) **Aug. 2026 – Sep. 2026**
Mechanical Engineering Intern *Ras Gharib, Egypt*

- Conducted CAD modeling and 2D drafting for mechanical components using SolidWorks and AutoCAD.
- Performed thermal and fluid dynamic simulations (CFD) using ANSYS Fluent to analyze pressure drop and heat transfer.
- Utilized Advanced Excel for engineering data analysis, regression models, and fluid performance calculations.
- Collaborated with cross-functional engineering teams during hands-on field training in oil field sites and power plants.
- Shadowed site engineers in the maintenance department, gaining direct insight into troubleshooting, equipment inspection, and mechanical repairs.

Projects

Reynolds Number Experiment – Fluid Flow Analysis | *Fluid Mechanics, Fluid Dynamics, Experimental Analysis, Data Processing*

- Conducted the Reynolds experiment to investigate and classify fluid flow behavior into laminar, transitional, and turbulent regimes.
- Utilized dye injection techniques in glass pipes across varying flow rates to visually observe and document flow transition patterns.
- Calculated volumetric flow rates, flow velocities, and Reynolds numbers using standard fluid mechanics governing equations.
- Validated experimental observations against theoretical predictions to confirm critical Reynolds number thresholds for pipe flow dynamics.

Smart Fire Truck – Mechanical Design & Automation | *SOLIDWORKS, 3D Printing, Arduino, Sensors, Mechatronics, System Testing*

- Designed and built a fully automated fire truck model, integrating mechanical, electrical, and software engineering principles.
- Developed the 3D mechanical structure and a specialized mechanical arm using SOLIDWORKS, followed by fabrication using 3D Printing technology.
- Programmed the vehicle using Arduino to incorporate multiple sensors for real-time fire source detection and temperature monitoring.
- Implemented an automatic suppression system and a custom interface for remote control, managing the full engineering cycle from concept to testing.

Double-Pipe Heat Exchanger CFD Study & Validation | *ANSYS Fluent, SolidWorks, CFD Simulation, Heat Transfer Analysis*

- Conducted a comprehensive CFD study on a Double-Pipe Heat Exchanger using ANSYS Fluent across multiple helical groove configurations.
- Modeled complex 3D geometry in SolidWorks and created high-quality structured meshing for accurate flow domain analysis.
- Analyzed fluid dynamics, thermal enhancement, pressure drop, and heat transfer efficiency at various Reynolds numbers.
- Validated numerical CFD simulation models against experimental data, achieving an accurate match with minimal margin of error.

Certifications & Training

- **SOLIDWORKS Mechanical Design Associate (CSWA)** – Dassault Systèmes 2026
- **SOLIDWORKS Additive Manufacturing Associate** – Dassault Systèmes 2026
- **SOLIDWORKS Sustainability Associate** – Dassault Systèmes 2026
- **SOLIDWORKS Electrical Design Associate** – Dassault Systèmes 2026
- **SOLIDWORKS xDesign Associate** – Dassault Systèmes 2026
- **SOLIDWORKS xShape Associate** – Dassault Systèmes 2026
- **SOLIDWORKS xMold Associate** – Dassault Systèmes 2026
- **Practical Field Engineering Internship** – General Petroleum Company (GPC) 2026
- **Computational Fluid Dynamics (CFD) Workshop** – Eng. Abdullah Hamdy 2026
- **HVAC Systems Design Training** – IT Training Center 2026
- **Data Analysis Using Microsoft Excel** – Edraak 2026
- **SOLIDWORKS Mechanical Design Certificate** – M3aarf Platform 2026